



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)

UNIT TEST

18-10-2020

JEE(Main + Advanced) : NURTURE COURSE (PHASE : TEPS)

ANSWER KEY

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	A	B	A	D	D	A	C,D	A	A,B,D
	Q.	11	12	13	14	15	16	17	18		
	A.	A,C,D	A,B,C,D	A,C,D	A,C	A	C	B	C		

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	A	D	B	D	B	A	A,B,D	A,B,D	A,B,C,D
	Q.	11	12	13	14	15	16	17	18		
	A.	A,B,C,D	A,B,D	A,C	B,D	B	D	B	C		

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	B	D	A	B	A	B	A,C	A,B,C,D	A,B,C,D
	Q.	11	12	13	14	15	16	17	18		
	A.	A,B	B,C,D	B,C	D	D	A	A	C		



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PART-2 : CHEMISTRY

SOLUTION

SECTION-I

1. Ans. (C)

2. Ans. (A)

3. Ans. (D)

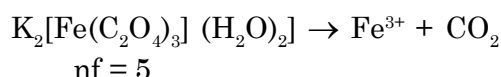
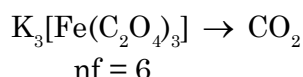
4. Ans. (B)

5. Ans. (D)

6. Ans. (B)

$$6x + 5x = 66 \times 0.2 \times 5 = 66$$

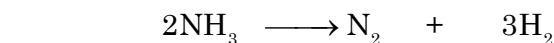
$$x = 6$$



Let the moles of each compound is = x

$$6x + 5x = 66 \times 0.2 \times 5 \Rightarrow x = 6$$

7. Ans. (A)

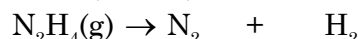


$$t = 0 \quad a \text{ mole} \quad b \text{ mole} \quad 0$$

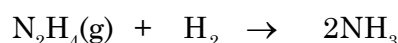
$$t = t \quad a - x \quad b + \frac{x}{2} \quad \frac{3x}{2}$$

$$\frac{b}{a+b} \quad \frac{b+\frac{x}{2}}{a+b+x} \Rightarrow \frac{b}{a} = \frac{1}{1}$$

8. Ans. (A,B,D)



$$1-x-y \quad x \quad 2x-y$$



$$1-x-y \quad 2x-y \quad 2y$$

$$\text{Total pressure} = 1 - x - y + x + 2x - y + 2y = 1.4$$

$$\Rightarrow 1 + 2x = 1.4$$

$$\Rightarrow x = 0.2$$

$$\Rightarrow P_{H_2} = 0.3 = 2x - y \Rightarrow y = 0.1 \text{ atm}$$

$$\Rightarrow K_{p1} = \frac{(0.2)(0.3)^2}{0.7} = \frac{18}{7} \times 10^{-2} (\text{atm})^2$$

$$\Rightarrow K_{p2} = \frac{(0.2)^2}{0.7 \times 0.3} = \frac{4}{21}$$

9. Ans. (A,B,D)

10. Ans. (A,B,C,D)

11. Ans. (A,B,C,D)

12. Ans. (A,B,D)

13. Ans. (A,C)

14. Ans. (B,D)

15. Ans. (B)

16. Ans. (D)

17. Ans. (B)

18. Ans. (C)

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. Ans. (B)

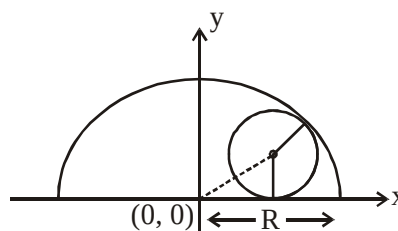
$$\text{Sol. } T_1 = \frac{1}{9}, \frac{T_2}{T_1} = \left(\frac{2}{5}\right)^2, \frac{T_3}{T_2} = \left(\frac{3}{7}\right)^2 \dots$$

$$\therefore \sum T_i < \frac{1/9}{1-1/4} = \frac{4}{27}$$

2. Ans. (B)

$$\text{Sol. } OP + y = R$$

$$\sqrt{x^2 + y^2} = R - y$$



$$\text{solving } x^2 = R^2 - 2Ry$$

Hence parabola

3. Ans. (D)

$$\text{Sol. } \left| \frac{PQ}{PR} \right| = 5$$

4. Ans. (A)

$$\text{Sol. } \left(\frac{x^2 + 2x + 2}{x+1} \right)^{2018} = a_{2018}x^{2018} + a_{2017}x^{2017} + \dots$$

$$+ a_1x + a_0 + \frac{b_1}{x+1} + \frac{b_2}{(x+1)^2} + \dots + \frac{b_{2018}}{(x+1)^{2018}}$$

$$\left(x+1 + \frac{1}{x+1} \right)^{2018} = {}^{2018}C_0(x+1)^{2018} + {}^{2018}C_1(x+1)^{2017}$$

$$+ {}^{2018}C_2(x+1)^{2016} + \dots + {}^{2018}C_{1008}(x+1)^2 + {}^{2018}C_{1009}$$

$$+ {}^{2018}C_{2018} \left(\frac{1}{x+1} \right)^{2018}$$

Put $x = 1$

$$\sum_{i=0}^{2018} a_i + \sum_{i=1}^{2018} \frac{b_i}{2^i} = \left(2 + \frac{1}{2} \right)^{2018} = \left(\frac{5}{2} \right)^{2018} \rightarrow (A)$$

Put $x = 0$,

$$2^{2018} = \left({}^{2018}C_0 + {}^{2018}C_1 + \dots + {}^{2018}C_{1008} + {}^{2018}C_{1009} \right)$$

$$+ \underbrace{\left({}^{2018}C_{1010} + {}^{2018}C_{1011} + \dots + {}^{2018}C_{2018} \right)}_{\sum_{i=1}^{2018} b_i}$$

$$\Sigma b_i = \frac{2^{2018} - {}^{2018}C_{1009}}{2}$$

$$a_0 = \frac{2^{2018} + {}^{2018}C_{1009}}{2}$$

5. Ans. (B)

$$\text{Sol. } \sum_{r=0}^{n-p} (-1)^{p+r} \frac{\binom{n}{r}}{\binom{n-r}{r}} \cdot \frac{\binom{n-r}{p}}{\binom{n-p-r}{p}} 5^{n-p-r} \cdot 6^r$$

$$= \frac{\binom{n}{p}}{\binom{n-p}{p}} \sum_{r=0}^{n-p} \frac{\binom{n-p}{r}}{\binom{n-p-r}{p}} (-1)^{p+r} 5^{n-p-r} 6^r$$

$$= (-1)^p \cdot {}^nC_p \sum_{r=0}^{n-p} {}^{n-p}C_r 5^{(n-p)-r} (-6)^r$$

$$= {}^nC_p (-1)^n$$

6. Ans. (A)

Sol. Do Your Self

7. Ans. (B)

Sol. Case-I:

Two rows contain 2 letters each and one row has 1 letter.

$$\text{Possible ways} = 4 \times {}^3C_2 + {}^4C_2 \times {}^3C_1 + {}^4C_2 \times {}^3C_2$$

$$\times 2 = 12 + 18 + 36 = 66$$

Case-II:

One row has 3 letters and two others have 1 letter each.

$$\text{Possible ways} = {}^4C_3 \times {}^3C_1 \times {}^2C_1 + {}^4C_1 \times {}^2C_1 = 4$$

$$\times 3 \times 2 + 4 \times 2 = 24 + 8 = 32$$

Hence total arrangements

$$= \frac{(66+32)5!}{2!} = 5880$$

8. Ans. (A,C)

$$\text{Sol. We know that } r_1r_2 + r_2r_3 + r_3r_1 = \frac{r_1r_2r_3}{r} = s^2$$

$$\therefore 4(r_2 + r_3) + r_2r_3 = 2r_2r_3 = 144$$

$$\Rightarrow r_2 + r_3 = 18 ; r_2r_3 = 72$$

$$\Rightarrow r_2 = 6 ; r_3 = 12 \quad (\text{or otherwise})$$

$$\therefore |r_2 - r_3| = 6$$

$$\text{Also, } \frac{r}{r_1} = \frac{s-a}{s} = \frac{1}{2} \Rightarrow a = 6$$

Similarly $b = 8 ; c = 10$ (or otherwise)

$\therefore \Delta ABC$ is right Δ .

$$\text{Area} = \frac{1}{2} \times 8 \times 6 = 24 \text{ sq. units}$$

$$\text{Least angle} = \sin^{-1} \left(\frac{3}{5} \right)$$

$$R = \frac{abc}{4\Delta} = \frac{6 \times 8 \times 10}{4(24)} = 5$$

9. Ans. (A,B,C,D)

$$\text{Sol. } \begin{vmatrix} 2 & 1 & -1 \\ a & 3 & -3 \\ 3 & 2 & -2 \end{vmatrix} = 0$$

on solving $(2a - 9) - (2a - 9) = 0$

Thus concurrent for all values of 'a'

10. Ans. (A,B,C,D)

$$\text{Sol. } (x^4 + y^4 + 2x^2y^2 - 2x^2 - 2y^2 + 1) - 4y^2 = 0$$

$$(x^2 + y^2 - 1)^2 - (2y)^2 = 0$$

$$(x^2 + y^2 - 2y - 1)(x^2 + y^2 + 2y - 1) = 0$$

$$\text{Circles are } x^2 + y^2 - 2y - 1 = 0$$

$$\text{and } x^2 + y^2 + 2y - 1 = 0$$

(do yourself)

11. Ans. (A,B)

Sol. $P_1 : y^2 = \ell_1 x$

$P_2 : y^2 = -\ell_2(x - b)$

point on $P_1 \left(\frac{y^2}{\ell_1}, y \right)$ and $P_2 \left(b - \frac{y^2}{\ell_2}, y \right)$

find locus of midpoint of above points

12. Ans. (B,C,D)

Sol. $\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$ touches $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at point $(x_1, -y_1)$. Also the locus of foot of perpendicular from the focus $(\sqrt{a^2 + b^2}, 0)$ to any tangent is the auxiliary circle $x^2 + y^2 = a^2$.

13. Ans. (B,C)

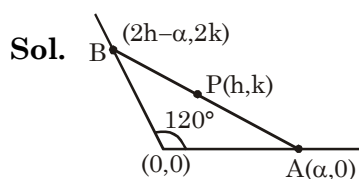
Sol. $\frac{1 - (1+x)^{32}}{1 - (1+x)} = \sum_{r=0}^{32} a_r x^r$

$\Rightarrow \frac{(1+x)^{32} - 1}{x} = {}^{32}C_1 + {}^{32}C_2 x + \dots + {}^{32}C_{32} x^{31} = \sum_{r=0}^{32} a_r x^{r-1}$

14. Ans. (D)

Sol. If we interchange any two rows of a determinant in set B, its value becomes -1 and hence it is in set C.

Paragraph for Question 15 and 16



Let $(\alpha, 0) \Rightarrow B(2h - \alpha, 2k)$

$\frac{2k}{2h - \alpha} = -\sqrt{3} \Rightarrow \alpha = \frac{2k + 2\sqrt{3}h}{\sqrt{3}}$

$4k^2 + (2h - 2\alpha)^2 = 100$

$3x^2 + 7y^2 + 4\sqrt{3}xy = 75$

Let $x = r \cos \theta$, $y = r \sin \theta$

$\Rightarrow r^2 = \frac{75}{3 \cos^2 \theta + 7 \sin^2 \theta + 4\sqrt{3} \sin \theta \cos \theta}$

$r^2 = \frac{75}{3 + 4 \sin^2 \theta + 2\sqrt{3} \sin 2\theta}$

$r^2 = \frac{75}{5 + (2\sqrt{3} \sin 2\theta - 2 \cos 2\theta)}$

$r_{\max}^2 = \frac{75}{5 + (-4)} = 75$

$\Rightarrow r_{\max} = 5\sqrt{3}$

15. Ans. (D)

16. Ans. (A)

Paragraph for Question 17 & 18

17. Ans. (A)

Sol. Number of ways $= {}^2C_1 \cdot ({}^2C_1 \cdot {}^{11}C_9 + {}^2C_2 \cdot {}^{11}C_8)$

18. Ans. (C)

Sol. A B C
Batsman Bowler Wicket keeper
Required number of ways

${}^{12}C_9 + 2 \cdot {}^{13}C_{10} + {}^{13}C_{11}$
(when A & B both play) (exactly one of A or B play) (A & B both do not play)

or

Number of ways ${}^{15}C_{11} - {}^{12}C_8$
(Total) (when A, B, C all play)